

A

B

C

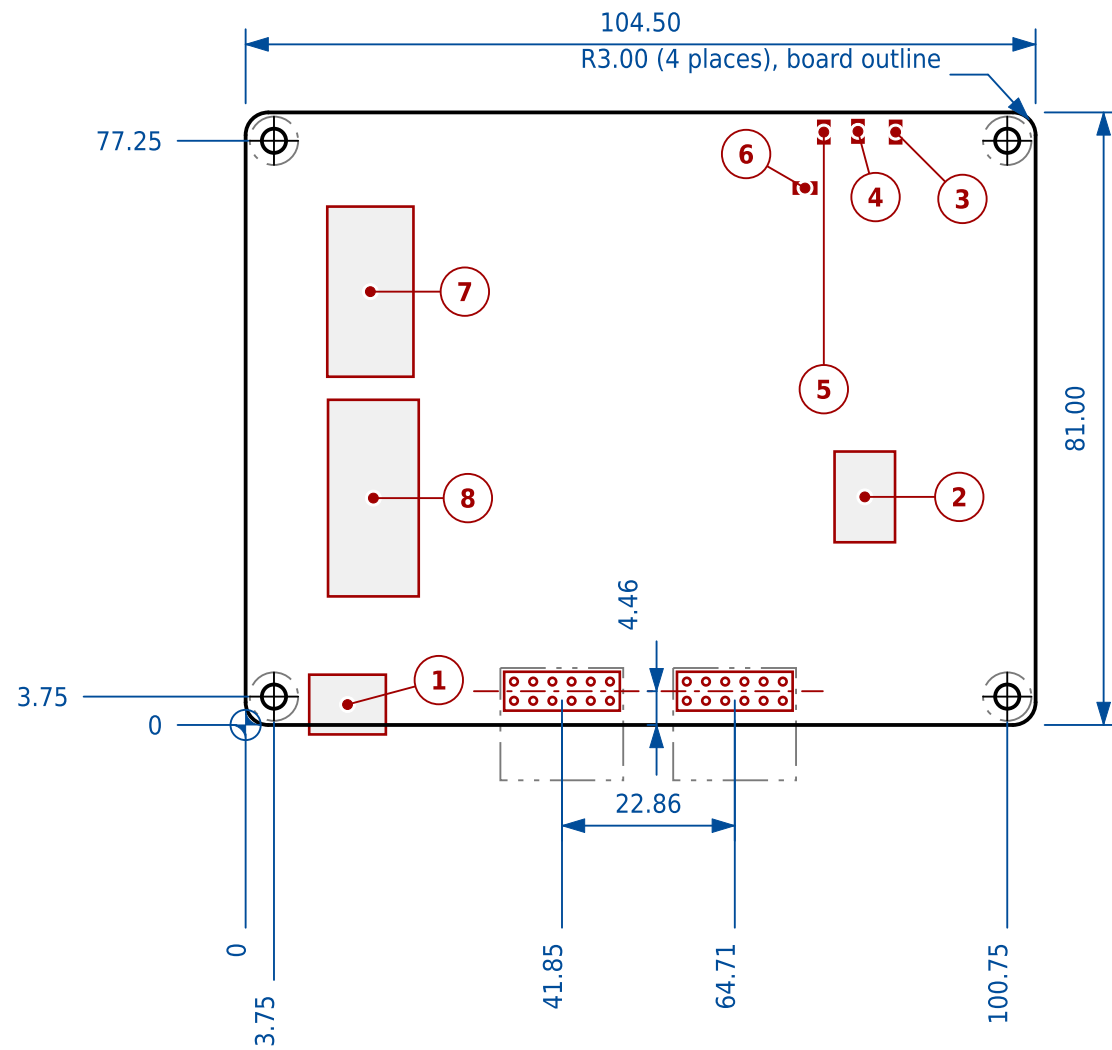
D

A

B

C

D



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- The title block reads "TinyTapeout 1,2,3 Demo Board", but no PCB served TT01: that was a bare-die trial run.
- This sheet covers revisions 2.2.5 and 2.2.6: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.60000 mm is its stackup sum, not a specified finished thickness.

NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 82.25 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt123-demo-pcb> mpw-mb1.kicad_pcb @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
- KiCad board file: <https://github.com/TinyTapeout/tt123-demo-pcb> mpw-mb1.kicad_pcb @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
- Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370 - used by: TT02, TT03

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
PMOD A	J3	bottom	41.850	4.465	48.200	5.735
PMOD B	J9	bottom	64.710	4.465	71.060	5.735

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	8.41 to 18.55	-1.25 to 6.65
2	7-segment display	77.91 to 85.91	24.16 to 36.16
3	LED D1	85.25 to 86.72	76.92 to 79.88
4	LED D2	80.27 to 81.73	77.02 to 79.98
5	LED D3	75.75 to 77.22	76.92 to 79.88
6	LED D4	72.52 to 75.48	70.27 to 71.73
7	8-way input DIP switch	10.80 to 22.20	46.05 to 68.55
8	9-way DIP switch SW2	10.90 to 22.90	17.00 to 43.00
9	Side Pmod, first position - not on this board	-	-
10	Side Pmod, second position - not on this board	-	-
11	Side Pmod, third position - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-88-g5151f5c+	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-01	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	